

Arthur Chien

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EDUCATION

Carnegie Mellon University, School of Computer Science

Pittsburgh, PA

Master of Science in Artificial Intelligence and Innovation(MSAII), GPA: 4.0/4.0

August 2024 – May 2026

National Chengchi University

Taipei, Taiwan

*Bachelor of Science in Management Information Systems, GPA: 4.0/4.3, Dean's List * 4 August 2019 – January 2024*

Coursework: Operating System, Data Structure, Algorithm, Computer Network, Database, Distributed Systems, Statistics, Discrete Mathematics, Linear Algebra, Machine Learning, Deep Learning, Natural Language Processing

SKILLS

Programming Languages: Python, Java, C/C++, SQL, JavaScript (Node.js), HTML/CSS, R, Bash, Shell Script

Machine Learning & Deep Learning: PyTorch, TensorFlow, Scikit-learn, Transformers, Computer Vision, NLP, Optimization Algorithms, Graph Neural Networks, LLMs, MLflow, LangGraph, LangChain

DevOps & Backend: Linux, Docker, Kubernetes, Apache Kafka, RESTful APIs, PostgreSQL, MongoDB, Redis, AWS, GCP, Azure, Git/GitHub, CI/CD (Actions, Jenkins), Prometheus, Grafana, OpenTelemetry, Elasticsearch, Kibana

WORKING EXPERIENCE

Machine Learning Engineer & Capstone Researcher

Pittsburgh, PA, USA

Bank of New York Mellon (BNY) & Carnegie Mellon University

September 2025 – May 2026

- Engineered a system for **self-evolving AI agents** using **React**, **OpenHands**, and **FastAPI**, integrating **multi-modal** skill ingestion to extract actionable tool schemas and workflows from video and audio instructions
- Built an agent reasoning loop with LLM-based **self-reflection** that increased task completion rates by **21%** through real-time failure recovery and iterative trajectory optimization
- Developed an evaluation framework using **Docker sandboxes** to benchmark agents in ground truth environments

Software Engineer Intern

Hsinchu, Taiwan

Taiwan Semiconductor Manufacturing Company (TSMC)

June 2025 – August 2025

- Developed automated test case generation from **OpenTelemetry** traces by constructing execution-flow graphs and analyzing call paths, achieving **88%** recall of failing services in pre-production.
- Parsed 1k+ flowchart XMLs into graphs and generated natural language docs with LLMs for a **RAG-based AI** assistant, improving documentation efficiency and reducing manual effort.

Research Assistant

Taipei, Taiwan

Data Mining and Machine Learning Laboratory, Academia Sinica

April 2023 – July 2023

- Built a document similarity module using **SimGNN** with **GCNs** and **SimHash**, improving scientific paper recommendations by **30%** and boosting link prediction accuracy by **12%** on a 1M+ document dataset

PROJECTS

Scalable Movie Recommendation System with Monitoring and Continuous Deployment

- Built a scalable movie recommendation system for **1M+ users** with **Kafka** for real-time ingestion, **MongoDB** for storage, and containerized **RESTful API** microservices using **Docker** and **Kubernetes**
- Automated model retraining and deployment every **48h** via **Jenkins** pipelines and versioned models in **MLflow**
- Monitored system metrics via **Prometheus** and **Grafana**, including click-through rate (CTR), latency, and alerts

Latent Guard++: Context-Aware Safety Framework for Generative Models

- Built a hybrid safety filtering system for **20k+ text-to-image prompts**, enhancing baseline Latent Guard with LLM-guided context awareness and tested across in-/out-of-distribution and adversarial datasets
- Proposed **dynamic thresholding** that adapts classification boundaries using **LLM-estimated risk confidence**
- Achieved **13.9%** accuracy improvement on **unseen datasets** and **10% gain on adversarial prompts**

Automated Hit-frame Detection for Badminton Match Analysis

- Implemented a **microservices**-based badminton analysis system using **PyTorch**, **OpenCV**, and **Flask**, processing **500+** professional match videos and reducing manual analysis time by **75%**, as presented in a [first-author paper](#)
- Architected a **CNN**-based rally segmentation and player tracking pipeline, achieving **89% precision** in rally identification and **97.5% PCK@0.2** for in-court player detection with a custom-trained **Keypoint-RCNN**
- Developed a seq2seq **Transformer**-based shuttle direction prediction model, achieving **96% accuracy**